MSG No. 108.238 I_c4cm [Type IV, tetragonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{4^+_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
3	$\{4^-_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
4	$\{2_{001}\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
5	$\{\mathrm{m}_{100}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
6	$\{\mathrm{m}_{010}\|00\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
7	$\{\mathrm{m}_{110}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
8	$\{\mathrm{m}_{1-10}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
9	$\{1\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
10	$\{4^+_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
11	$\{4^-_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
12	$\{2_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
13	$\{\mathrm{m}_{100}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{010}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	1
15	$\{\mathrm{m}_{110}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0 & -1 & 0 & \frac{1}{2}\\ -1 & 0 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	1
16	$\{\mathrm{m}_{1-10}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0 & 1 & 0 & \frac{1}{2}\\ 1 & 0 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	1
17	$\{1'\vert00\frac{1}{2}\}$	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
18	$\{4^+_{001}'\vert00\frac{1}{2}\}$	$\begin{bmatrix}0 & -1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
19	$\{4^-_{001}'\vert00\frac{1}{2}\}$	$\begin{bmatrix}0 & 1 & 0 & 0\\ -1 & 0 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
20	$\{2_{001}'\vert00\frac{1}{2}\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$	1	-1
21	$\{\mathrm{m}_{100}'\vert0\}$	$\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	-1
22	$\{\mathrm{m}_{010}'\vert0\}$	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	-1
23	$\{\mathrm{m}_{110}'\vert0\}$	$\begin{bmatrix}0 & -1 & 0 & 0\\ -1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	-1
24	$\{\mathrm{m}_{1-10}'\vert0\}$	$\begin{bmatrix}0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	-1	-1
25	$\{1'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	1	-1
26	$\{4^+_{001}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0 & -1 & 0 & \frac{1}{2}\\ 1 & 0 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	1	-1
27	$\{4^-_{001}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0 & 1 & 0 & \frac{1}{2}\\ -1 & 0 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & 0\end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{2_{001}'\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	-1
29	$\{\mathrm{m}_{100}'\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
30	$\{\mathrm{m}_{010}'\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
31	$\{\mathrm{m}_{110}'\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1
32	$\{\mathrm{m}_{1-10}'\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	-1